Chapter 9
Wildfire Spread Simulation

Forest fires, whether naturally occurring or prescribed, are potential risks for ecosystems and human settlements. These risks can be managed by monitoring the weather, prescribing fires to limit available fuel, and creating firebreaks. With computer simulations we can predict and explore how fires may spread. We can explore scenarios and test different fire management strategies under different weather conditions. Using Tangible Landscape and the GRASS GIS wildfire toolset we simulated several wildfire scenarios. We tested different configurations of firebreaks on the physical model and evaluated their effectiveness.

9.1 Fire Spread Modeling Methods

Fire spread across landscapes is a complex, highly dynamic process influenced by weather, topography, and fuel. The fire spread models predict the rate of fire spread and fire perimeter growth using a combination of physical principles such as energy conservation and empirical parameters derived from observations and experimental data. For reviews of physical and empirical models of fire spread and methods of implementing fire simulation please refer to Sullivan (2009a,b,c).

9.1.1 Input Data

The fuel model is one of the most important input variables for fire simulation. Albini (1976) and Anderson (1982) describe 13 classes of fuel (Table 9.1), which differ in fuel loads and the distribution of fuel particle size classes. The size of individual pieces of fuel influences the fire; heat is absorbed faster by small twigs

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Table 9.1 The fuel model by Anderson (1982) describes 13 classes of fuel

Fuel class	Description
Grass and grass-dominated	
1	Short grass (1 foot)
2	Timber (grass and understory)
3	Tall grass (2.5 feet)
Chaparral and shrub fields	
4	Chaparral (6 feet)
5	Brush (2 feet)
6	Dormant brush, hardwood slash
7	Southern rough
Timber litter	
8	Closed timber litter
9	Hardwood litter
10	Timber (litter and understory)
Slash	
11	Light logging slash
12	Medium logging slash
13	Heavy logging slash

due to their large surface to volume ratio. The moisture content of the fuel affects the amount of heat needed to ignite the fuel. The smaller the fuel size, the faster the fuel dries out and becomes more combustible.

The slope of the terrain and the speed and direction of the wind control the rate of spread and direction of the fire. Steeper slopes cause faster ignition in the upslope direction. This rule has been observed and quantified in several laboratory experiments (Silvani et al. 2012; Viegas 2004; Weise and Biging 1996).

Wind is specified by speed and direction at a given height. Unlike the slope of the terrain, wind is difficult to characterize due to its variability in space and time and spatially and temporally averaged values are used in the models. Sparks and embers can be carried great distances by strong winds; this process known as spotting allows fire to spread beyond firebreaks (Albini 1983).

9.1.2 Fire Spread Algorithm

In our simulation we used the GRASS GIS modules *r.ros* and *r.spread*, implemented by Xu (1994), to compute the rate of spread and simulate the spread of fire.

The rate of spread computation is based on the BEHAVE model (Andrews 1986). The inputs are the fuel model, fuel moisture, wind speed and direction and terrain slope and aspect. The rate of spread is computed using the following equation:

$$R = \frac{I_R \xi (1 + \Phi_W + \Phi_S)}{\rho_b \varepsilon Q_{ig}} \tag{9.1}$$

where:

R	is the rate of spread in m/s
I_R	is the reaction intensity in kW/m
ξ	is the propagating flux ratio
Φ_W	is the wind coefficient
Φ_S	is the slope factor
ρ_b	is the oven-dry fuel per cubic meter of fuel bed in kg/m^3
ε	is the effective heating number
Q_{ig}	is the heat of preignition in kJ/kg.

For a detailed description of the underlying mathematical model and an explanation of the input data refer to Rothermel (1972).

The module *r.spread* uses Huygens' principle to simulate elliptically anisotropic spread where each cell center is a potential origin of spread and the local spread is ellipsoidal (Anderson et al. 1982). The sizes and orientations of the ellipses vary by cell. Local wind and slope directions determine the orientation of ellipses. The module *r.spread* uses a specific implementation of the shortest path algorithm.

We used the GRASS GIS add-on module *r.fire.spread* to call the modules *r.ros* and *r.spread* streamlining the fire spread computation and visualization. With this add-on we can use temporally variable input conditions and export the fire spread for each time step so that we can animate the simulation with the module *g.gui.animation*.

9.2 Case Study: Controlling Fire with Firebreaks

With Tangible Landscape we can intuitively create new fire management scenarios, rapidly testing how different sizes, shapes, and placements of firebreaks affect the spread of fire. To test different management strategies we simulated how a fire might spread across NCSU's Centennial Campus and tried to contain it by creating firebreaks of different sizes and alignments. In this case study we modeled a phenomenon—fire spread—that started outside the extent of the physical model, but spread towards the model (Fig. 9.1). Once the fire spread onto the model we could interact with it by creating firebreaks. Geospatial models and simulations need not be constrained to the boundary of the physical model, i.e. the extent of the tangible user interface. The processing extent can extend beyond the interactive area allowing us to work across scales.

We built a malleable physical model of the tree canopy by casting polymeric sand. To build the molds for casting we interpolated a DEM and a DSM from 2013 lidar data and then CNC routed the DEM and the inverse of the DSM. Then we cast a

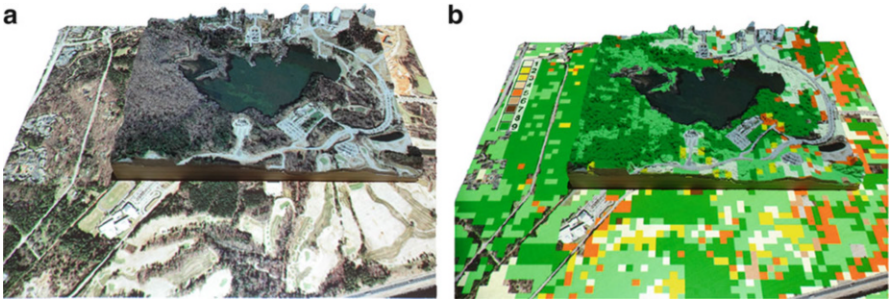

Fig. 9.1 The study area on Centennial Campus: (**a**) an orthophotograph and (**b**) available fuel map projected over the table with the tangible 3D model

layer of sand representing the tree canopy between the two molds. Since this layer of sand represents tree canopy and thus fuel availability we could model clearcutting simply by removing sand. Thus we were able to intuitively design firebreaks by carving into the sand layer and reducing the fuel load. Please refer to Sect. 6.2.1 for a more detailed description of the site and model.

9.2.1 Data Preparation

To simulate fire spread we prepared several input raster layers for the module *r.ros* including the data about terrain, wind, fuel, and moisture conditions. We also specified the coordinates of location where the fire started.

Terrain Terrain slope influences the speed of spread and aspect influences the spread direction. After we set the region to our study area we derived the slope and aspect raster maps from the provided DEM:

```
g.region n=224134 s=222501 e=639326 w=637211 res=3
r.slope.aspect elevation=dem slope=slope aspect=aspect
```

The DEM is also used as input for computing the maximum spotting distance.

Wind Wind has two components—midflame velocity and direction—and both can be spatially variable. In this case we used the prevailing wind speed and direction obtained from the nearest weather station through the State Climate Office of North Carolina.[1] The module *r.ros* requires velocity in feet per minute so the wind velocity data acquired from the State Climate Office of North Carolina must be converted from meters per second to feet per minute (to convert multiply by a factor of approximately 197). Wind direction is typically reported by the direction *from* which

[1] State Climate Office of North Carolina: www.nc-climate.ncsu.edu/

it originates, clockwise from the north. The module *r.ros*, however, requires the "to" direction (also clockwise from north). To simulate spatial variability we applied a random effect, for example, using the module *r.surf.gauss* to produce a raster map of Gaussian deviates with a specified mean and standard deviation:

```
r.surf.gauss output=wind_speed_avg mean=542 sigma=30
r.surf.gauss output=wind_dir mean=75 sigma=20
```

Fuel The module *r.ros* uses Anderson's 13 standard fire behavior fuel models. Fuel data layers at 30 m resolution for the USA are publicly accessible via the LANDFIRE website.[2] Most of our study site falls into fuel classes 8 and 9, i.e. timber litter (Fig. 9.1b).

Moisture Since fuel moisture data were not readily available for our site we generated the required, spatially variable raster maps of 1-h fuel moisture and live fuel moisture percentage for dry conditions:

```
r.surf.gauss output=moisture_1h mean=10 sigma=5
r.surf.gauss output=moisture_live mean=20 sigma=5
```

Starting Location The starting sources of the fire are represented as raster cells and can be created by digitizing points or importing coordinates from a file and converting them to a raster. We provided the fire starting location in a text file with the point coordinates (for example `638097,222934`) and converted them to a raster with the module *v.to.rast*:

```
v.in.ascii input=source.txt output=source separator=comma
v.to.rast input=source output=source type=point use=cat
```

9.2.2 Scenario with Multiple Firebreaks

In our first scenario the fire started near Trailwood Drive and was drawn to the northeast by a southwesterly wind. The fire jumped over the road and quickly spread towards the Chancellor's House (Fig. 9.2).

Using the following workflow we simulated and then animated the spread of the fire. The output raster maps generated by this workflow represent the relative time of ignition in minutes. To better visually represent the spread of the fire we used a red-yellow-gray color ramp. The steepness of the color gradient shows the relative rate of spread. The steeper the color gradient, the slower the spread. To prepare the suggested color ramp we saved the color rules in a plain text file `fire_colors.txt`:

[2]LANDFIRE: http://www.landfire.gov/

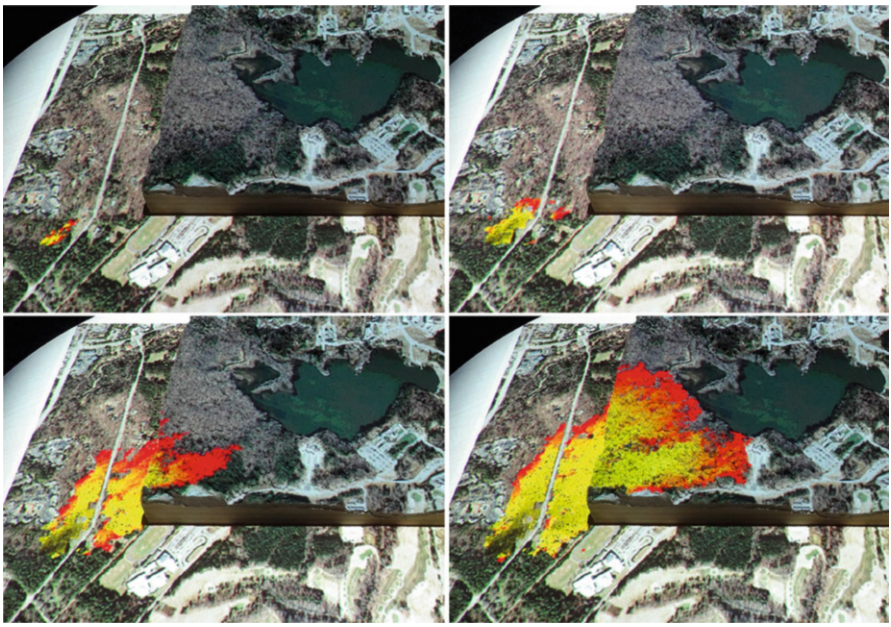

Fig. 9.2 The spread of the fire without intervention

```
0% 50:50:50
60% yellow
100% red
```

To run the simulation we called the module *r.ros* once and then *r.spread* with the desired time lag parameter specifying the length of the simulation. To see the intermediate states we can run the module *r.spread* multiple times and assign our color ramp to the resulting spread raster map:

```
r.ros model=fuel moisture_1h=moisture_1h \
    moisture_live=moisture_live velocity=wind_speed_avg \
    direction=wind_dir slope=slope aspect=aspect \
    elevation=elevation base_ros=out_base_ros \
    max_ros=out_max_ros direction_ros=out_dir_ros \
    spotting_distance=out_spotting

r.spread -s base_ros=out_dir_ros max_ros=out_max_ros \
    direction_ros=out_dir_ros start=source \
    spotting_distance=out_spotting wind_speed=wind_speed_avg \
    fuel_moisture=moisture_1h output=spread_1 lag=20
r.spread -s -i base_ros=out_dir_ros max_ros=out_max_ros \
    direction_ros=out_dir_ros start=spread_1 \
    spotting_distance=out_spotting wind_speed=wind_speed_avg \
    fuel_moisture=moisture_1h output=spread_2 lag=20
r.spread ...
```

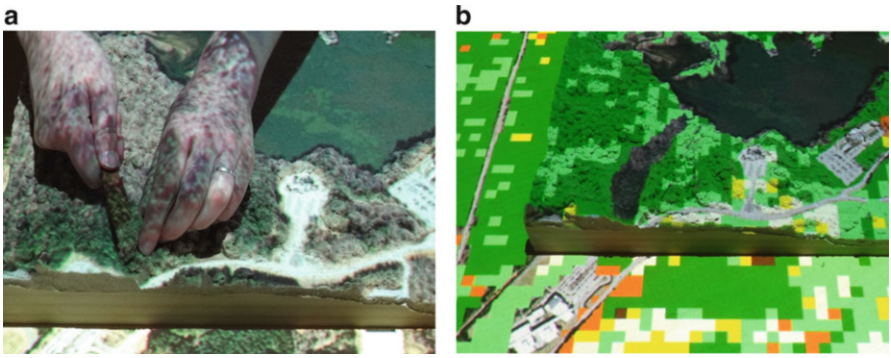

Fig. 9.3 Creating a firebreak by (**a**) manually removing sand and then (**b**) scanning and detecting the change

```
r.null map=spread_1 setnull=0
r.colors map=spread_1 rules=fire_colors.txt
```

Alternatively we can use the add-on module *r.fire.spread*, which conveniently wraps the previous sequence of commands into a single command:

```
r.fire.spread -s start=source times=0 end_time=1600 \
    time_step=20 output=spread model=fuel \
    moisture_1h=moisture_1h moisture_live=moisture_live \
    direction=wind_dir slope=slope aspect=aspect \
    elevation=elevation speed=wind_speed_avg
```

After simulating the initial spread of the fire we attempted to prevent fire from spreading towards Chancellor's House by creating firebreaks. First we scanned the model to save the unmodified state. Next we manually removed sand (representing canopy) from the location where we want to have a firebreak (see Fig. 9.3a). We scanned the modified model and vertically matched the new scan to the unmodified scan using the function defined in code snippet 4.2.2. We derived the new fuel raster layer based on the difference between the two scans introducing no data values into the copy of the original fuel model using a simple raster algebra expression:

```
adjust_scan('scan_before', 'scan_after', 'scan_adjusted')
gscript.mapcalc("changed_fuel = if(scan_before - scan_adjusted
    > 0, null(), fuel)")
```

We repeated our simulation using the new fuel raster layer. The resulting fire spread in Fig. 9.4 shows that our attempt was only a partial success. While we significantly slowed the spread of the fire, potentially giving firefighters more time to act, the fire jumped over the firebreak and started to spread towards Chancellor's House suggesting that a wider firebreak would be needed. With more precise data about fire behavior we could control and limit the spotting effect in the simulation.

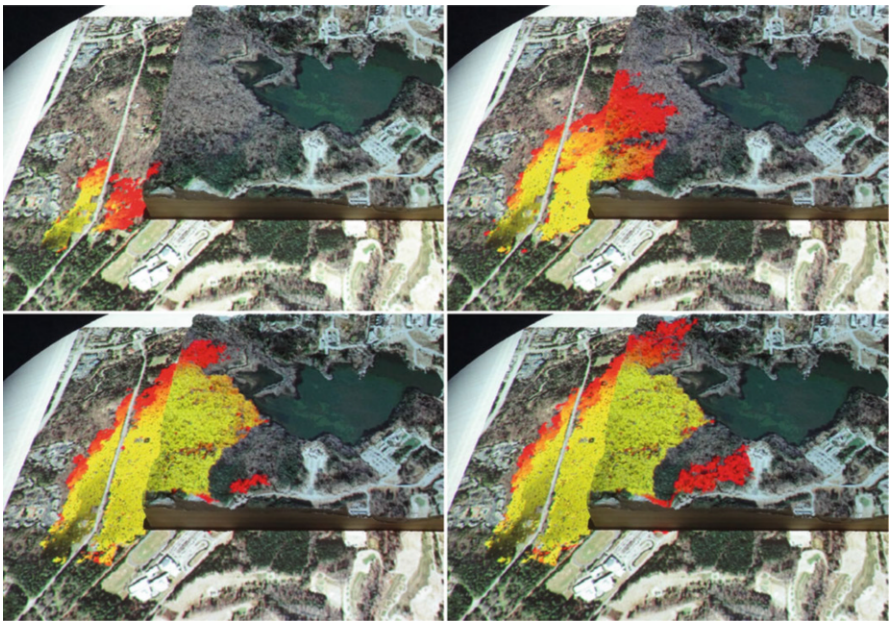

Fig. 9.4 The spread of the fire after creating a firebreak

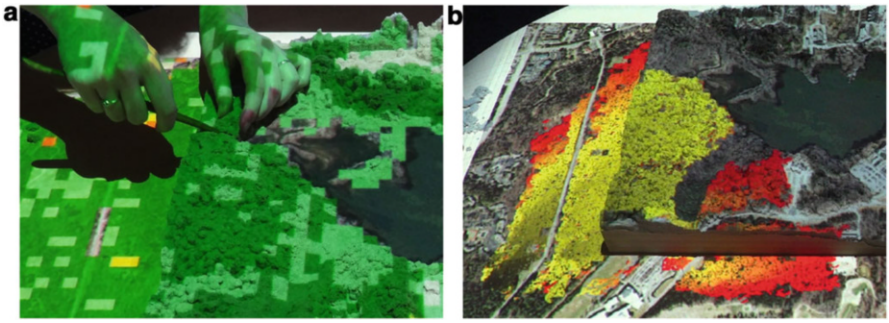

Fig. 9.5 The simulation with additional firebreak: (**a**) the creation of a new firebreak by removing sand and (**b**) the result of the new simulation

In our second scenario we added another firebreak to protect the neighborhood north of Lake Raleigh. The lake creates a natural barrier that constrains the spread of the fire. To protect the community to the north we extended this barrier by creating an additional firebreak starting on western shore of the lake (Fig. 9.5a). After we rescanned the model and reran the simulation, we found that the additional firebreak was only locally effective—while the fire was significantly slowed, it would have eventually reached the neighborhood.

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